

Problem 1: Parametric Tangent Lines

Problem: Given the parametric curve $x(t) = 2 \cos(3t) - 3 \sin(2t)$ and $y(t) = 4 \sin(3t) + 3 \cos(4t)$. Find the equations for tangent lines at $t = 0$ and $t = \pi$.

Solution

First, we find the derivatives of x and y with respect to t to determine the slope $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt}[2 \cos(3t) - 3 \sin(2t)] = -6 \sin(3t) - 6 \cos(2t) \\ \frac{dy}{dt} &= \frac{d}{dt}[4 \sin(3t) + 3 \cos(4t)] = 12 \cos(3t) - 12 \sin(4t)\end{aligned}$$

Case 1: At $t = 0$

1. **Find the point (x_0, y_0) :**

$$x(0) = 2 \cos(0) - 3 \sin(0) = 2(1) - 0 = 2$$

$$y(0) = 4 \sin(0) + 3 \cos(0) = 0 + 3(1) = 3$$

Point: $(2, 3)$.

2. **Find the slope m :**

$$x'(0) = -6 \sin(0) - 6 \cos(0) = -6$$

$$y'(0) = 12 \cos(0) - 12 \sin(0) = 12$$

$$m = \frac{12}{-6} = -2$$

3. **Equation of the line:**

$$y - 3 = -2(x - 2) \implies y = -2x + 4 + 3 \implies \boxed{y = -2x + 7}$$

Case 2: At $t = \pi$

1. **Find the point (x_1, y_1) :**

$$x(\pi) = 2 \cos(3\pi) - 3 \sin(2\pi) = 2(-1) - 0 = -2$$

$$y(\pi) = 4 \sin(3\pi) + 3 \cos(4\pi) = 0 + 3(1) = 3$$

Point: $(-2, 3)$.

2. **Find the slope m :**

$$x'(\pi) = -6 \sin(3\pi) - 6 \cos(2\pi) = 0 - 6(1) = -6$$

$$y'(\pi) = 12 \cos(3\pi) - 12 \sin(4\pi) = 12(-1) - 0 = -12$$

$$m = \frac{-12}{-6} = 2$$

3. **Equation of the line:**

$$y - 3 = 2(x - (-2)) \implies y = 2x + 7 \implies \boxed{y = 2x + 7}$$

Problem 2: Area Perimeter and Surface Area

Problem: Region R bounded by $y = x^2$ and $y = 4 - 2x - x^2$. (i) Find the perimeter. (ii) Find the surface area of the solid obtained by rotating R around the x -axis.

Solution

Intersection Points: Set $x^2 = 4 - 2x - x^2 \implies 2x^2 + 2x - 4 = 0 \implies x^2 + x - 2 = 0 \implies (x + 2)(x - 1) = 0$. The curves intersect at $x = -2$ and $x = 1$. The upper curve is $y_T = 4 - 2x - x^2$ and the lower curve is $y_B = x^2$.

(i) Perimeter Evaluation

The perimeter is the sum of the arc lengths of the top and bottom curves. Formula:
 $L = \int_a^b \sqrt{1 + (y')^2} dx$.

- Top curve (y_T): $y' = -2 - 2x = -2(1 + x)$.
- Bottom curve (y_B): $y' = 2x$.

$$P = \int_{-2}^1 \sqrt{1 + 4(1 + x)^2} dx + \int_{-2}^1 \sqrt{1 + 4x^2} dx$$

Using the standard integral $\int \sqrt{1 + u^2} du = \frac{u}{2} \sqrt{1 + u^2} + \frac{1}{2} \operatorname{arsinh}(u)$: Let $u = 2(1 + x)$ for the first integral ($du = 2dx$) and $u = 2x$ for the second ($du = 2dx$). The integral becomes:

$$\int \sqrt{1 + u^2} \frac{du}{2} = \frac{1}{4} \left(u \sqrt{1 + u^2} + \operatorname{arsinh}(u) \right)$$

Step-by-step Evaluation:

$$1. \text{ Top Curve Integral: } I_1 = \left[\frac{2(x+1)}{4} \sqrt{1 + 4(x+1)^2} + \frac{1}{4} \operatorname{arsinh}(2(x+1)) \right]_{-2}^1$$

- At $x = 1$: $\frac{4}{4} \sqrt{1 + 16} + \frac{1}{4} \operatorname{arsinh}(4) = \sqrt{17} + \frac{1}{4} \operatorname{arsinh}(4)$.
- At $x = -2$: $\frac{-2}{4} \sqrt{1 + 4} + \frac{1}{4} \operatorname{arsinh}(-2) = -\frac{\sqrt{5}}{2} - \frac{1}{4} \operatorname{arsinh}(2)$.
- $I_1 = (\sqrt{17} + \frac{1}{4} \operatorname{arsinh}(4)) - (-\frac{\sqrt{5}}{2} - \frac{1}{4} \operatorname{arsinh}(2)) = \sqrt{17} + \frac{\sqrt{5}}{2} + \frac{1}{4} \operatorname{arsinh}(4) + \frac{1}{4} \operatorname{arsinh}(2)$.

$$2. \text{ Bottom Curve Integral: } I_2 = \left[\frac{2x}{4} \sqrt{1 + 4x^2} + \frac{1}{4} \operatorname{arsinh}(2x) \right]_{-2}^1$$

- At $x = 1$: $\frac{2}{4} \sqrt{1 + 4} + \frac{1}{4} \operatorname{arsinh}(2) = \frac{\sqrt{5}}{2} + \frac{1}{4} \operatorname{arsinh}(2)$.
- At $x = -2$: $\frac{-4}{4} \sqrt{1 + 16} + \frac{1}{4} \operatorname{arsinh}(-4) = -\sqrt{17} - \frac{1}{4} \operatorname{arsinh}(4)$.
- $I_2 = (\frac{\sqrt{5}}{2} + \frac{1}{4} \operatorname{arsinh}(2)) - (-\sqrt{17} - \frac{1}{4} \operatorname{arsinh}(4)) = \frac{\sqrt{5}}{2} + \sqrt{17} + \frac{1}{4} \operatorname{arsinh}(2) + \frac{1}{4} \operatorname{arsinh}(4)$.

Total Perimeter:

$$P = I_1 + I_2 = 2\sqrt{17} + \sqrt{5} + \frac{1}{2} \operatorname{arsinh}(4) + \frac{1}{2} \operatorname{arsinh}(2)$$

Using $\operatorname{arsinh}(x) = \ln(x + \sqrt{x^2 + 1})$:

$$P = 2\sqrt{17} + \sqrt{5} + \frac{1}{2} \ln(4 + \sqrt{17}) + \frac{1}{2} \ln(2 + \sqrt{5})$$

(ii) Surface Area (Rotation about x-axis)

1. Calculating S_{bottom} The bottom curve is $y = x^2$. The derivative is $y' = 2x$.

$$S_{bottom} = \int_{-2}^1 2\pi(x^2)\sqrt{1+(2x)^2} dx = 2\pi \int_{-2}^1 x^2\sqrt{1+4x^2} dx$$

We use the standard integral formula for $\int u^2\sqrt{1+4u^2}du$:

$$\int x^2\sqrt{1+4x^2} dx = \frac{x(1+8x^2)}{32}\sqrt{1+4x^2} - \frac{1}{64}\ln\left(2x + \sqrt{1+4x^2}\right)$$

(Derivation note: This comes from hyperbolic substitution $2x = \sinh t$ or integration by parts).

Now, evaluate from -2 to 1 :

- At $x = 1$: $\sqrt{1+4(1)} = \sqrt{5}$.

$$F(1) = \frac{1(9)}{32}\sqrt{5} - \frac{1}{64}\ln(2 + \sqrt{5}) = \frac{9\sqrt{5}}{32} - \frac{1}{64}\ln(2 + \sqrt{5})$$

- At $x = -2$: $\sqrt{1+4(4)} = \sqrt{17}$.

$$F(-2) = \frac{-2(33)}{32}\sqrt{17} - \frac{1}{64}\ln(-4 + \sqrt{17})$$

Using the identity $\ln(-4 + \sqrt{17}) = \ln\left(\frac{1}{4+\sqrt{17}}\right) = -\ln(4 + \sqrt{17})$:

$$F(-2) = -\frac{33\sqrt{17}}{16} + \frac{1}{64}\ln(4 + \sqrt{17})$$

Thus, the definite integral value is:

$$\int_{-2}^1 x^2\sqrt{1+4x^2} dx = \left(\frac{9\sqrt{5}}{32} + \frac{33\sqrt{17}}{16}\right) - \frac{1}{64}\left[\ln(2 + \sqrt{5}) + \ln(4 + \sqrt{17})\right]$$

Multiplying by 2π :

$$S_{bottom} = \pi \left[\frac{9\sqrt{5}}{16} + \frac{33\sqrt{17}}{8} - \frac{1}{32}\ln\left((2 + \sqrt{5})(4 + \sqrt{17})\right) \right] \quad (1)$$

2. Calculating S_{top} The top curve is $y = 4 - 2x - x^2$. The derivative is $y' = -2 - 2x = -2(1+x)$.

$$S_{top} = \int_{-2}^1 2\pi(4 - 2x - x^2)\sqrt{1+(-2(1+x))^2} dx$$

Substitute $u = 1 + x$, so $dx = du$. Limits: $x = -2 \rightarrow u = -1$; $x = 1 \rightarrow u = 2$. The polynomial transforms as: $4 - 2x - x^2 = 5 - (1+x)^2 = 5 - u^2$.

$$S_{top} = 2\pi \int_{-1}^2 (5 - u^2)\sqrt{1+4u^2} du$$

This splits into two integrals:

$$S_{top} = 2\pi \left[5 \int_{-1}^2 \sqrt{1+4u^2} du - \int_{-1}^2 u^2 \sqrt{1+4u^2} du \right]$$

Let $A(u) = \frac{u}{2}\sqrt{1+4u^2} + \frac{1}{4}\ln(2u + \sqrt{1+4u^2})$ be the antiderivative of $\sqrt{1+4u^2}$. Let $B(u)$ be the antiderivative of $u^2\sqrt{1+4u^2}$ (same form as in S_{bottom}).

We combine the terms: The integrand is $(5-u^2)\sqrt{1+4u^2}$. The combined antiderivative is:

$$G(u) = \frac{u(79-8u^2)}{32}\sqrt{1+4u^2} + \frac{81}{64}\ln(2u + \sqrt{1+4u^2})$$

Evaluate limits $[-1, 2]$:

- At $u = 2$: $\sqrt{17}$.

$$G(2) = \frac{2(79-32)}{32}\sqrt{17} + \frac{81}{64}\ln(4 + \sqrt{17}) = \frac{47\sqrt{17}}{16} + \frac{81}{64}\ln(4 + \sqrt{17})$$

- At $u = -1$: $\sqrt{5}$.

$$G(-1) = \frac{-1(79-8)}{32}\sqrt{5} + \frac{81}{64}\ln(-2 + \sqrt{5}) = -\frac{71\sqrt{5}}{32} - \frac{81}{64}\ln(2 + \sqrt{5})$$

Subtracting $G(2) - G(-1)$:

$$\int = \left(\frac{47\sqrt{17}}{16} + \frac{71\sqrt{5}}{32} \right) + \frac{81}{64} \left[\ln(4 + \sqrt{17}) + \ln(2 + \sqrt{5}) \right]$$

Multiplying by 2π :

$$S_{top} = \pi \left[\frac{47\sqrt{17}}{8} + \frac{71\sqrt{5}}{16} + \frac{81}{32} \ln \left((2 + \sqrt{5})(4 + \sqrt{17}) \right) \right] \quad (2)$$

3. Total Surface Area Adding (1) and (2):

- $\sqrt{17}$ terms: $\frac{33}{8}\pi + \frac{47}{8}\pi = \frac{80}{8}\pi = 10\pi\sqrt{17}$.
- $\sqrt{5}$ terms: $\frac{9}{16}\pi + \frac{71}{16}\pi = \frac{80}{16}\pi = 5\pi\sqrt{5}$.
- Logarithmic terms: $-\frac{1}{32}\pi(\dots) + \frac{81}{32}\pi(\dots) = \frac{80}{32}\pi(\dots) = \frac{5}{2}\pi\ln(\dots)$.

Final Answer:
$$S = 5\pi \left(2\sqrt{17} + \sqrt{5} + \frac{1}{2} \ln \left[(2 + \sqrt{5})(4 + \sqrt{17}) \right] \right)$$

Problem 3: Taylor Series

Problem: Find the Taylor series for $f(x) = \sum_{k \geq 0} (\frac{3}{5^k} + (-1)^k \frac{5}{3^k}) x^k$ centered at $x = 1$.

Solution

First, simplify the given function by recognizing it as a sum of two geometric series $\sum ar^k = \frac{a}{1-r}$.

$$f(x) = 3 \sum_{k \geq 0} \left(\frac{x}{5}\right)^k + 5 \sum_{k \geq 0} \left(-\frac{x}{3}\right)^k$$

$$f(x) = \frac{3}{1 - \frac{x}{5}} + \frac{5}{1 - (-\frac{x}{3})} = \frac{15}{5 - x} + \frac{15}{3 + x}$$

To find the Taylor series centered at $c = 1$, we introduce a shift $t = x - 1 \implies x = t + 1$. We will expand in terms of t .

$$f(t+1) = \frac{15}{5 - (t+1)} + \frac{15}{3 + (t+1)} = \frac{15}{4 - t} + \frac{15}{4 + t}$$

Combine the fractions:

$$f(x) = 15 \left[\frac{(4+t) + (4-t)}{(4-t)(4+t)} \right] = 15 \left[\frac{8}{16 - t^2} \right] = \frac{120}{16 - t^2}$$

Now, express this as a geometric series in t^2 :

$$f(x) = \frac{120}{16(1 - \frac{t^2}{16})} = \frac{120}{16} \sum_{n=0}^{\infty} \left(\frac{t^2}{16}\right)^n = \frac{15}{2} \sum_{n=0}^{\infty} \frac{t^{2n}}{16^n}$$

Substitute $t = x - 1$ back:

$$f(x) = \frac{15}{2} \sum_{n=0}^{\infty} \frac{(x-1)^{2n}}{16^n}$$

Problem 4: Series Evaluation

Problem: Given $f(x) = \frac{x}{\log(1-x)}$, evaluate $\sum_{n \geq 1} \frac{x^n(1-x)^n}{n!} f^{(n)}(x)$.

Solution

This series resembles the Taylor series expansion formula:

$$f(x+h) = f(x) + \sum_{n=1}^{\infty} \frac{f^{(n)}(x)}{n!} h^n$$

By comparing the given series to the formula, we can identify $h = x(1-x)$. Thus, the sum S is:

$$S = f(x+h) - f(x)$$

Substitute $h = x - x^2$:

$$x+h = x + (x - x^2) = 2x - x^2$$

Now we evaluate the function f at the new point $2x - x^2$:

$$f(2x - x^2) = \frac{2x - x^2}{\log(1 - (2x - x^2))} = \frac{x(2 - x)}{\log(1 - 2x + x^2)} = \frac{x(2 - x)}{\log((1 - x)^2)}$$

Using the log property $\log(a^b) = b \log a$:

$$f(2x - x^2) = \frac{x(2 - x)}{2 \log(1 - x)}$$

Now compute $S = f(2x - x^2) - f(x)$:

$$S = \frac{2x - x^2}{2 \log(1 - x)} - \frac{x}{\log(1 - x)}$$

Find a common denominator:

$$S = \frac{(2x - x^2) - 2(x)}{2 \log(1 - x)} = \frac{2x - x^2 - 2x}{2 \log(1 - x)} = \frac{-x^2}{2 \log(1 - x)}$$

$$S = -\frac{x^2}{2 \log(1 - x)}$$

Problem 5: Polar Area

Problem: Find the area inside $r = 1 + \cos \theta$ (cardioid) and outside $r = \sqrt{3} \sin \theta$ (circle).

Solution

1. Intersection Points: $1 + \cos \theta = \sqrt{3} \sin \theta$. Using harmonic addition or inspection: At $\theta = \pi/3$: $1 + 1/2 = 3/2$ and $\sqrt{3}(\sqrt{3}/2) = 3/2$. Match. The curves also meet at the pole ($\theta = \pi$).

2. Region Analysis: We need the area **inside** the cardioid and **outside** the circle.

- For $\theta \in [0, \pi/3]$, the Cardioid is larger than the Circle. This is part of our region.
- For $\theta \in [\pi, 2\pi]$, the Circle is not defined (or geometric "negative" radius), so the Cardioid area counts entirely.

3. Integral Setup: Area = (Area between Cardioid and Circle in $[0, \pi/3]$) + (Area of Cardioid in $[\pi, 2\pi]$).

$$A = \frac{1}{2} \int_0^{\pi/3} \left[(1 + \cos \theta)^2 - (\sqrt{3} \sin \theta)^2 \right] d\theta + \frac{1}{2} \int_{\pi}^{2\pi} (1 + \cos \theta)^2 d\theta$$

4. Evaluation: Integral 1: $\int_0^{\pi/3} (1 + 2 \cos \theta + \cos^2 \theta - 3 \sin^2 \theta) d\theta$. Using identities: $3 \sin^2 \theta = 3(1 - \cos^2 \theta)$. Integrand = $4 \cos^2 \theta + 2 \cos \theta - 2 = 2 \cos 2\theta + 2 \cos \theta$.

$$\frac{1}{2} \int_0^{\pi/3} (2 \cos 2\theta + 2 \cos \theta) d\theta = \left[\frac{1}{2} \sin 2\theta + \sin \theta \right]_0^{\pi/3} = \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$$

Integral 2: $\frac{1}{2} \int_{\pi}^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta$.

$$A_2 = \frac{1}{2} \left[\theta + 2 \sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{\pi}^{2\pi} = \frac{1}{2} \left(3\pi - \frac{3\pi}{2} \right) = \frac{3\pi}{4}$$

Total Area:

$$A = \frac{3\sqrt{3}}{4} + \frac{3\pi}{4}$$

Problem 6: Radius and Interval of Convergence

Problem: $\sum_{n \geq 1} \frac{2^n(x-10)^n}{n^{1/3}}$.

(i) Radius of Convergence (ROC)

Use the Ratio Test on $a_n = \frac{2^n(x-10)^n}{n^{1/3}}$.

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}(x-10)^{n+1}}{(n+1)^{1/3}} \cdot \frac{n^{1/3}}{2^n(x-10)^n} \right| = 2|x-10|$$

For convergence, $L < 1$:

$$2|x-10| < 1 \implies |x-10| < \frac{1}{2}$$

$$\text{ROC} = \frac{1}{2}$$

(ii) Interval of Convergence (IOC)

Center is 10. Radius is 0.5. Endpoints are 9.5 and 10.5.

- **At $x = 10.5$ ($x - 10 = 1/2$):** Series becomes $\sum \frac{1}{n^{1/3}}$. p -series with $p = 1/3 \leq 1$. **Diverges.**
- **At $x = 9.5$ ($x - 10 = -1/2$):** Series becomes $\sum \frac{(-1)^n}{n^{1/3}}$. Alternating Series Test: $\frac{1}{n^{1/3}} \rightarrow 0$. **Converges.**

$$\text{IOC} = [9.5, 10.5)$$

Problem 7: Definite Integral

Problem: Evaluate $\int_0^2 \arcsin\left(\frac{2x}{1+x^2}\right) dx$.

Solution

We use the identity for $\arcsin\left(\frac{2x}{1+x^2}\right)$:

$$\arcsin\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2 \arctan x & \text{if } 0 \leq x \leq 1 \\ \pi - 2 \arctan x & \text{if } x > 1 \end{cases}$$

Therefore, we split the integral at $x = 1$:

$$I = \int_0^1 2 \arctan x \, dx + \int_1^2 (\pi - 2 \arctan x) \, dx$$

Compute Antiderivative: $\int \arctan x \, dx = x \arctan x - \frac{1}{2} \ln(1+x^2)$.

Evaluate Part 1:

$$\int_0^1 2 \arctan x \, dx = 2 \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) = \frac{\pi}{2} - \ln 2$$

Evaluate Part 2:

$$\begin{aligned} \int_1^2 (\pi - 2 \arctan x) \, dx &= \pi(2-1) - 2 \left[(2 \arctan 2 - \frac{1}{2} \ln 5) - (\frac{\pi}{4} - \frac{1}{2} \ln 2) \right] \\ &= \pi - 4 \arctan 2 + \ln 5 + \frac{\pi}{2} - \ln 2 = \frac{3\pi}{2} - 4 \arctan 2 + \ln(5/2) \end{aligned}$$

Total Integral:

$$I = \left(\frac{\pi}{2} - \ln 2 \right) + \left(\frac{3\pi}{2} - 4 \arctan 2 + \ln 5 - \ln 2 \right)$$

$$I = 2\pi - 4 \arctan 2 + \ln 5 - 2 \ln 2 = \boxed{2\pi - 4 \arctan 2 + \ln(5/4)}$$

Problem 8: Rational Antiderivative

Problem: Find the antiderivative of $f(x) = \frac{x^4+4x^3+11x^2+12x+8}{(x^2+2x+3)^2(x+1)}$.

Solution

1. Decomposition: Expand $(x^2 + 2x + 3)^2 = x^4 + 4x^3 + 10x^2 + 12x + 9$. Compare to numerator $N(x) = x^4 + 4x^3 + 11x^2 + 12x + 8$:

$$N(x) = (x^2 + 2x + 3)^2 + (x^2 - 1) = (x^2 + 2x + 3)^2 + (x - 1)(x + 1)$$

Substitute back:

$$f(x) = \frac{(x^2 + 2x + 3)^2}{(x^2 + 2x + 3)^2(x + 1)} + \frac{(x - 1)(x + 1)}{(x^2 + 2x + 3)^2(x + 1)} = \frac{1}{x + 1} + \frac{x - 1}{(x^2 + 2x + 3)^2}$$

2. Integration:

$$\int f(x)dx = \ln|x + 1| + \int \frac{x - 1}{(x^2 + 2x + 3)^2}dx$$

For the second integral, let $u = x + 1 \implies x - 1 = u - 2$. Denominator is $(u^2 + 2)^2$.

$$\int \frac{u - 2}{(u^2 + 2)^2}du = \int \frac{u}{(u^2 + 2)^2}du - 2 \int \frac{1}{(u^2 + 2)^2}du$$

- Integral A: $\int u(u^2 + 2)^{-2}du = -\frac{1}{2(u^2 + 2)}$.
- Integral B: $-2 \int \frac{1}{(u^2 + 2)^2}du$. Using $u = \sqrt{2} \tan \theta$, this yields $-2 \left(\frac{1}{4\sqrt{2}} \arctan\left(\frac{u}{\sqrt{2}}\right) + \frac{u}{4(u^2 + 2)} \right)$.

Final Result:

$$\boxed{\ln|x + 1| - \frac{x + 2}{2(x^2 + 2x + 3)} - \frac{\sqrt{2}}{4} \arctan\left(\frac{x + 1}{\sqrt{2}}\right) + C}$$